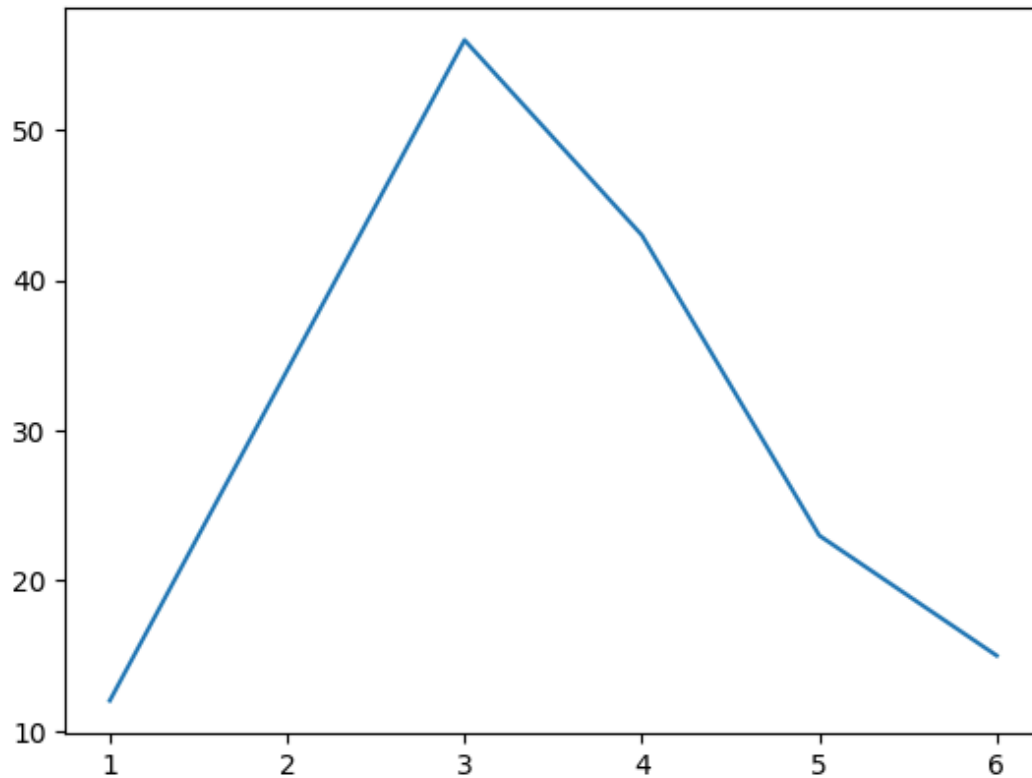


```
import matplotlib.pyplot as plt
%matplotlib inline
```

```
x = [1,2,3,4,5,6]
y = [12,34,56,43,23,15]
```

```
plt.plot(x,y)
```

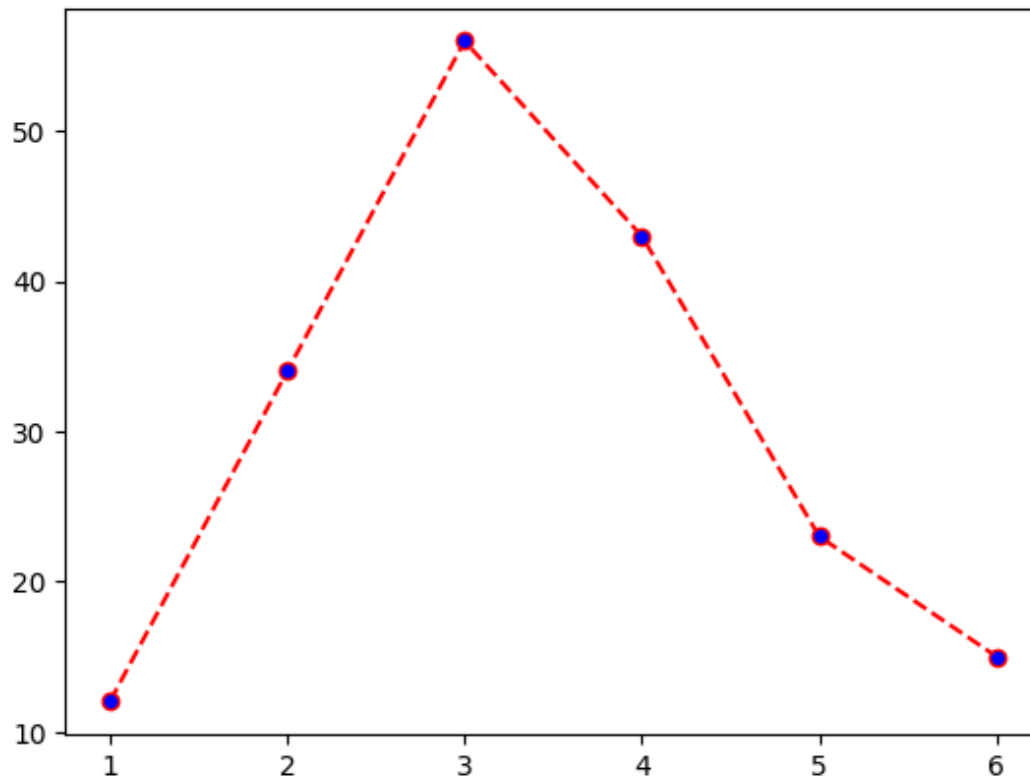
↗ [<matplotlib.lines.Line2D at 0x7f4851901a90>]



```
x = [1,2,3,4,5,6]
y = [12,34,56,43,23,15]
```

```
plt.plot(x,y, color='red',linestyle='dashed',marker='o',markersize=6,markerfacecolor='blue')
```

↗ [matplotlib.lines.Line2D at 0x7f48407d6e10>]

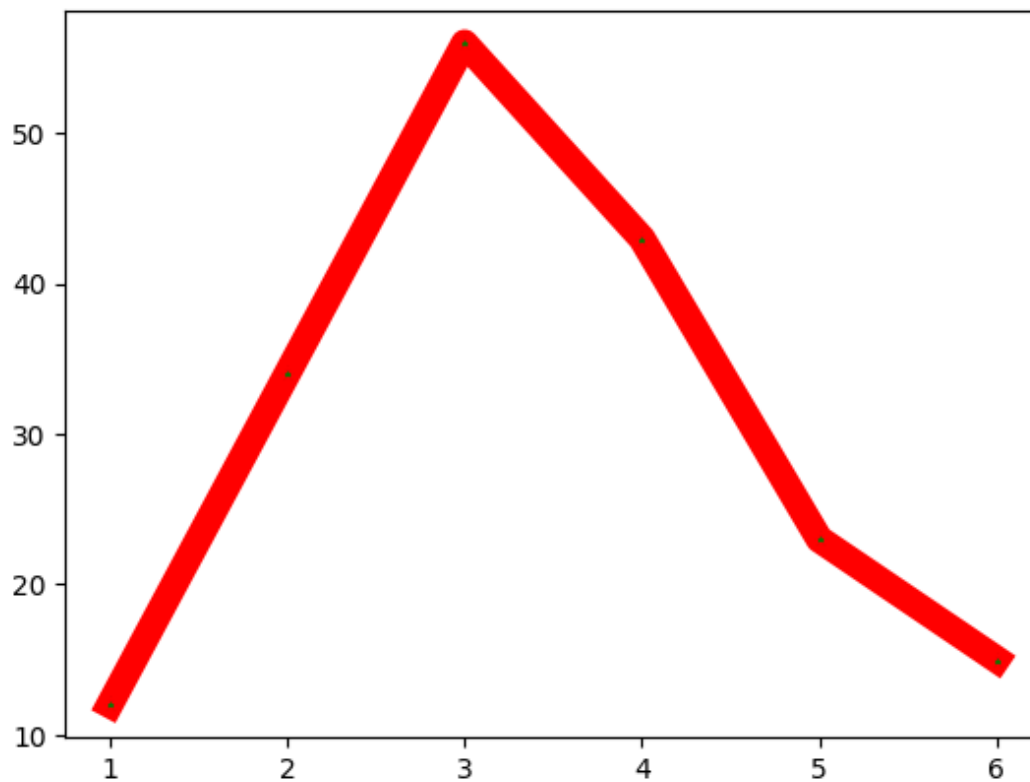


```
x = [1,2,3,4,5,6]
```

```
y = [12,34,56,43,23,15]          ### linestyle used (solid,dashed,dashdot,dotted,None,'')
```

```
plt.plot(x,y, 'r',linestyle='solid',marker='*',linewidth=10,markerfacecolor='green')
```

↗ [matplotlib.lines.Line2D at 0x7f4829d3fa10>]

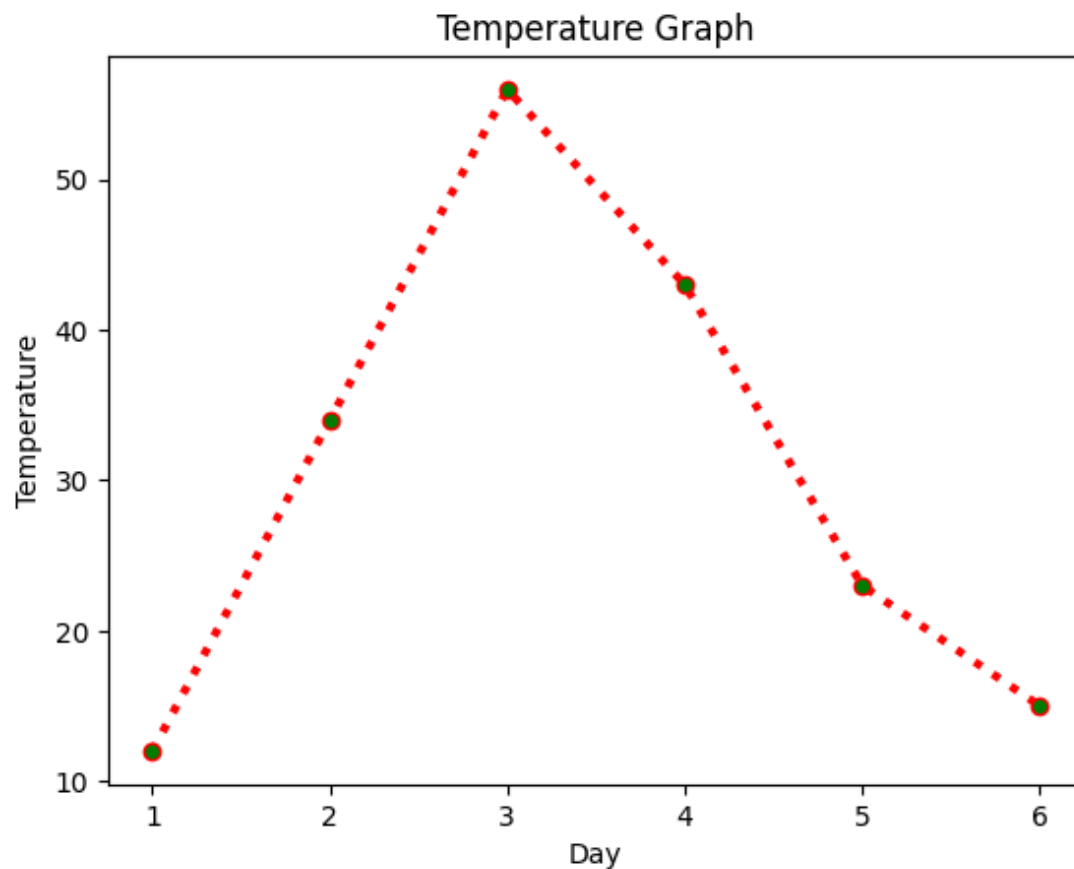


```
x = [1,2,3,4,5,6]
```

```
y = [12,34,56,43,23,15]
```

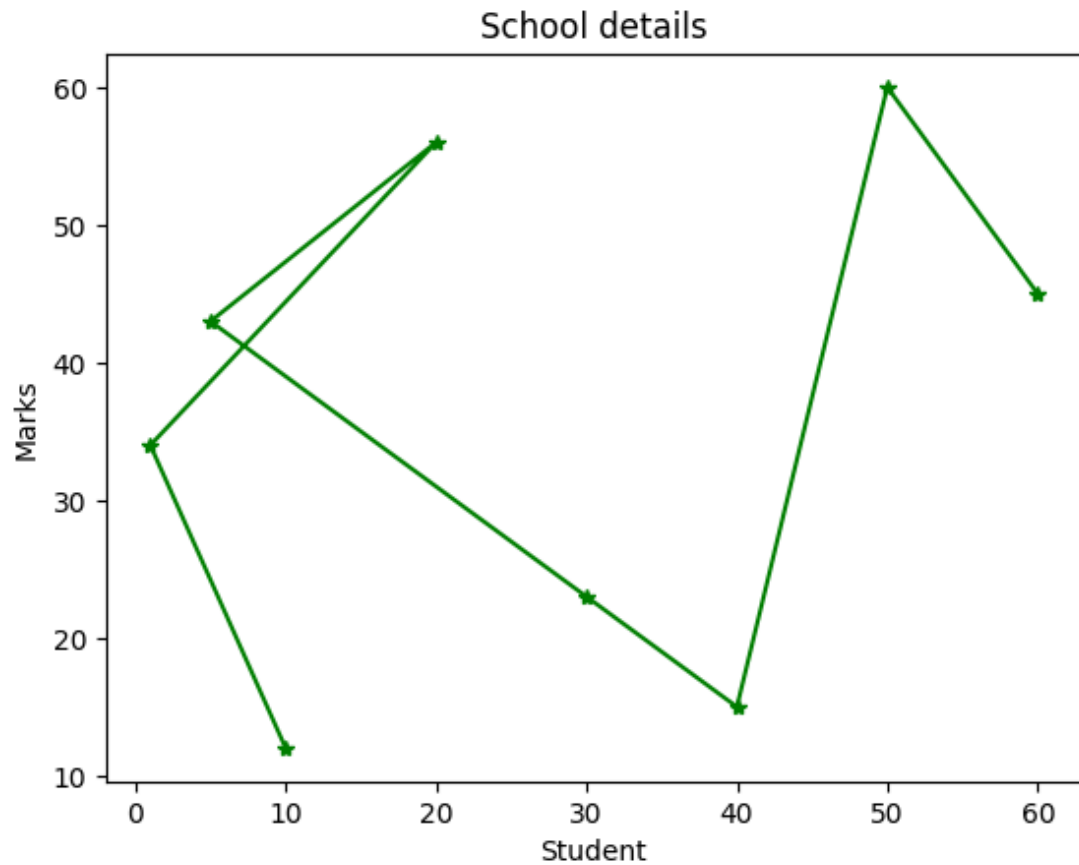
```
plt.xlabel('Day')
plt.ylabel('Temperature')
plt.title('Temperature Graph')
plt.plot(x,y, 'r',marker='o',linewidth = 3,linestyle='dotted',markerfacecolor='green')
plt.plot()
```

↗ []



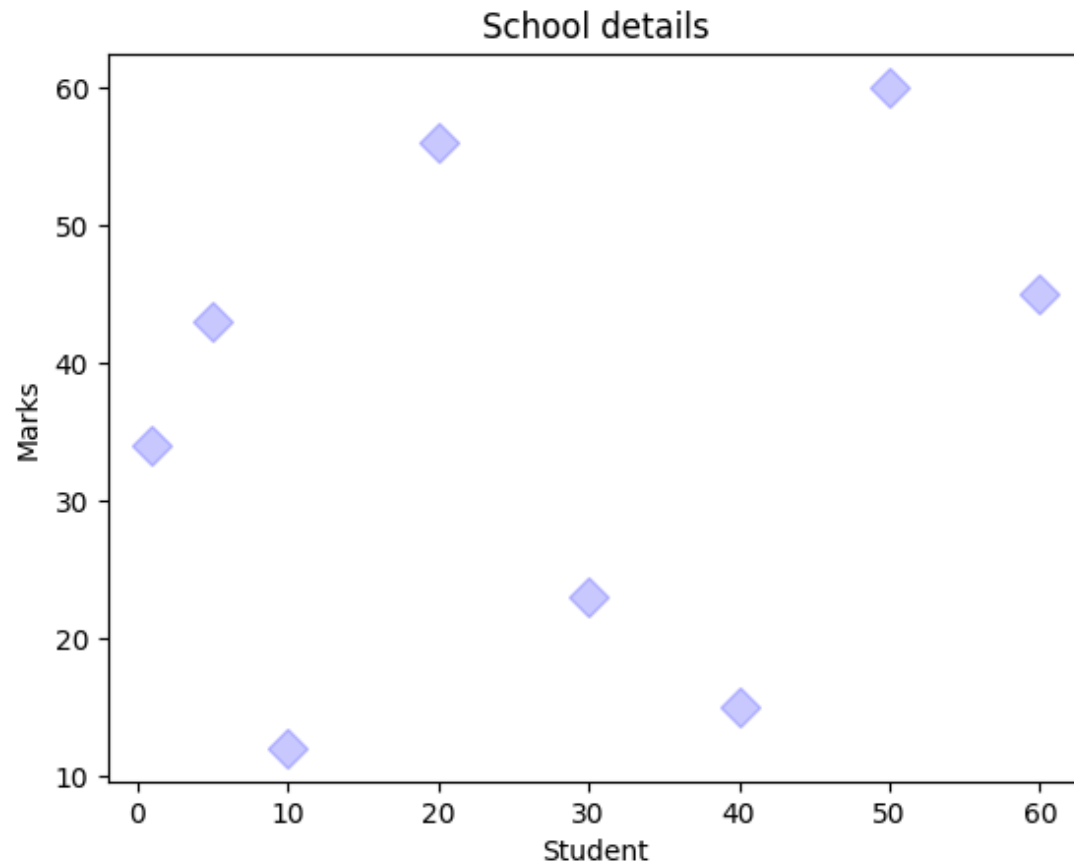
```
x = [10,1,20,5,30,40,50,60]
y = [12,34,56,43,23,15,60,45]
```

```
plt.xlabel('Student')
plt.ylabel('Marks')
plt.title('School details')
plt.plot(x,y, 'g*-',)
plt.plot()
```



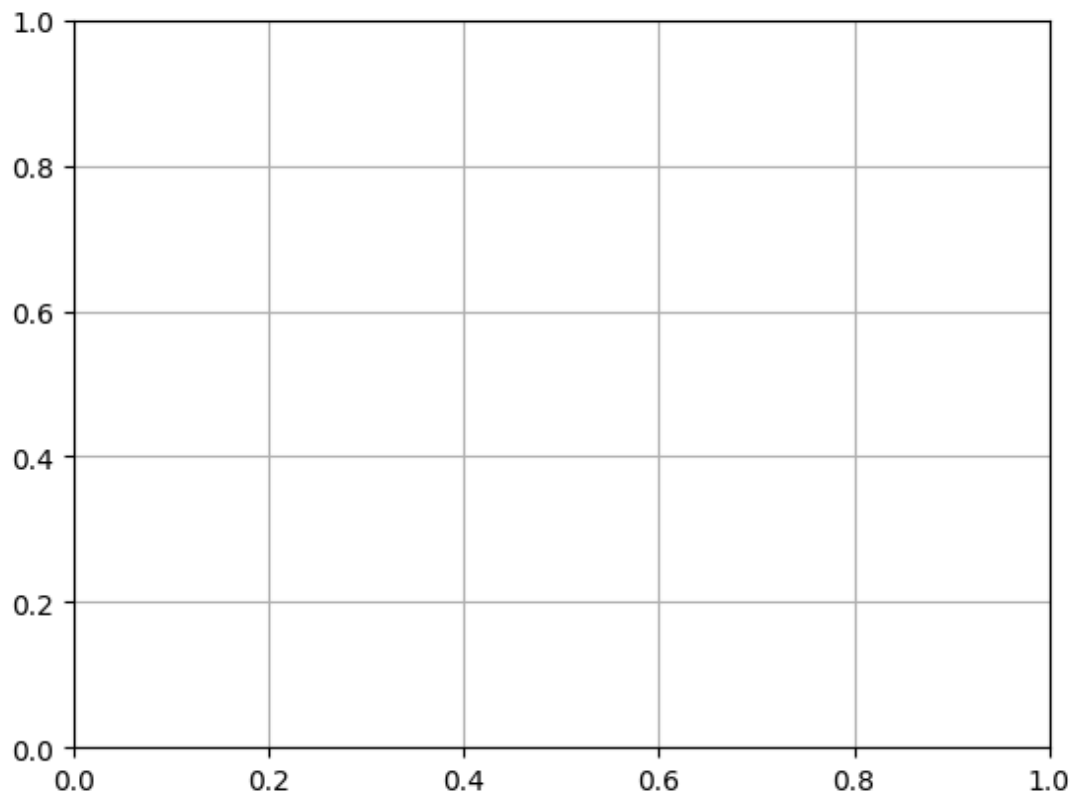
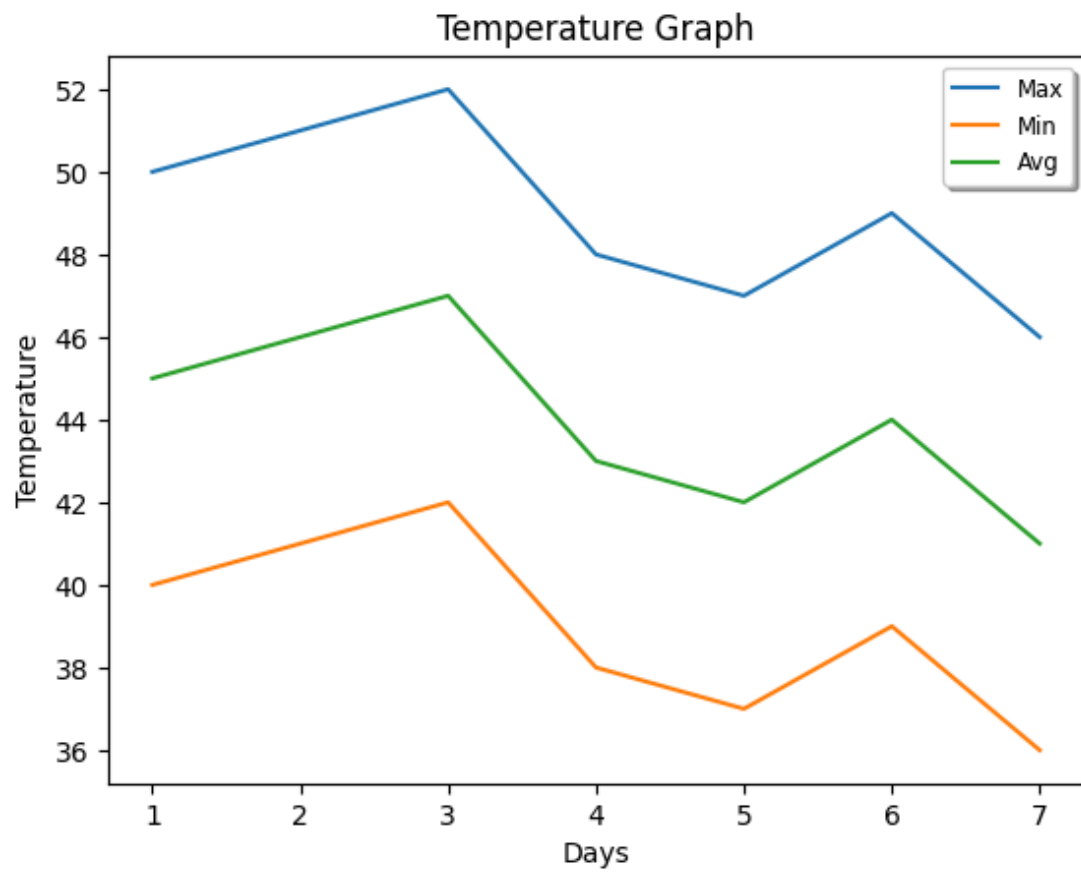
```
x = [10,1,20,5,30,40,50,60]
y = [12,34,56,43,23,15,60,45]
```

```
plt.xlabel('Student')
plt.ylabel('Marks')
plt.title('School details')
plt.plot(x,y, color='blue', marker='D', linestyle='', markersize=10, alpha=0.2)
plt.plot()
```



```
import matplotlib.pyplot as plt
# %matplotlib inline
days = [1,2,3,4,5,6,7]
max_t = [50,51,52,48,47,49,46]
min_t = [40,41,42,38,37,39,36]
avg_t = [45,46,47,43,42,44,41]

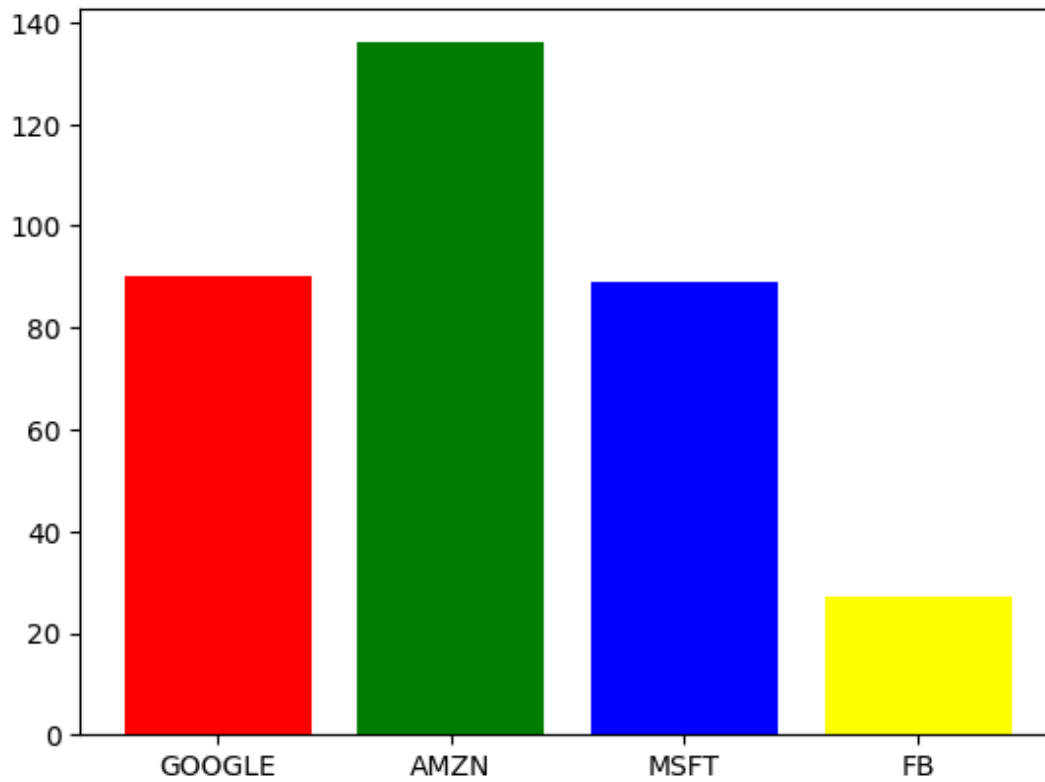
plt.xlabel('Days')
plt.ylabel('Temperature')
plt.title('Temperature Graph')
plt.plot(days,max_t, label="Max")
plt.plot(days,min_t, label="Min")
plt.plot(days,avg_t, label="Avg")
plt.legend(loc='upper left')
plt.legend(loc='best',shadow=True,fontsize='small')
# plt.legend(loc='center left',bbox_to_anchor=(1,0.5))
plt.show()
plt.grid()
```



Bar Graph

```
company = ['GOOGLE', 'AMZN', 'MSFT', 'FB']  
revenue = [90,136,89,27]  
profit = [12,45,68,55]  
plt.bar(company,revenue,color=['red','green','blue','yellow'])
```

↩️ <BarContainer object of 4 artists>



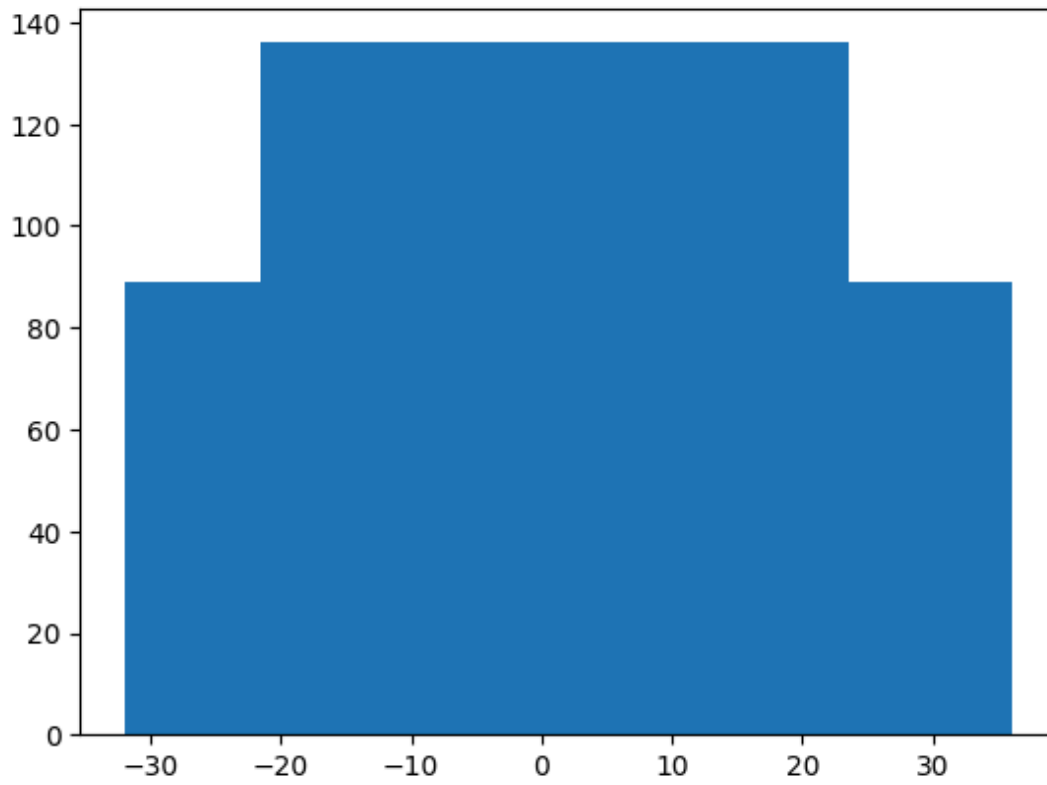
```
import numpy as np
company = ['GOOGLE', 'AMZN', 'MSFT', 'FB']
revenue = [90,136,89,27]
profit = [57, 44, 40, 21, 29]
```

```
x_pos = np.arange(len(company))
x_pos
```

↩️ array([0, 1, 2, 3])

```
plt.bar(x_pos,revenue,profit)
```

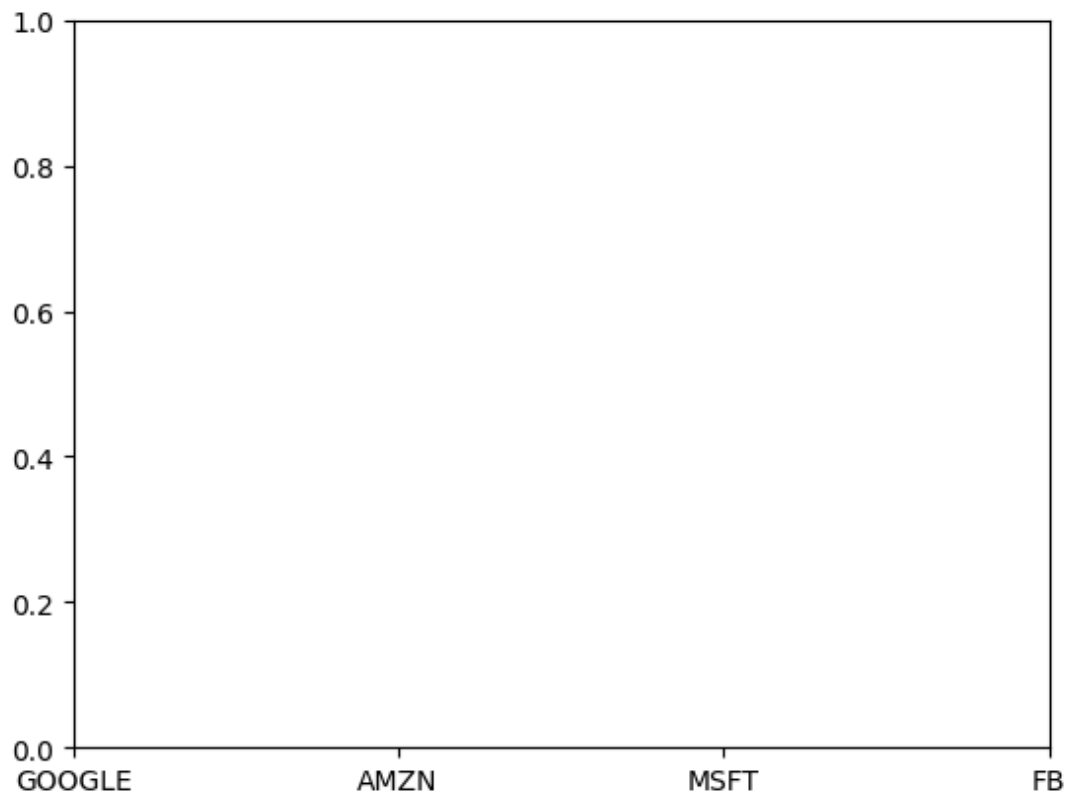
↔ <BarContainer object of 4 artists>



```
plt.xticks(x_pos,company) # show x - string val
```

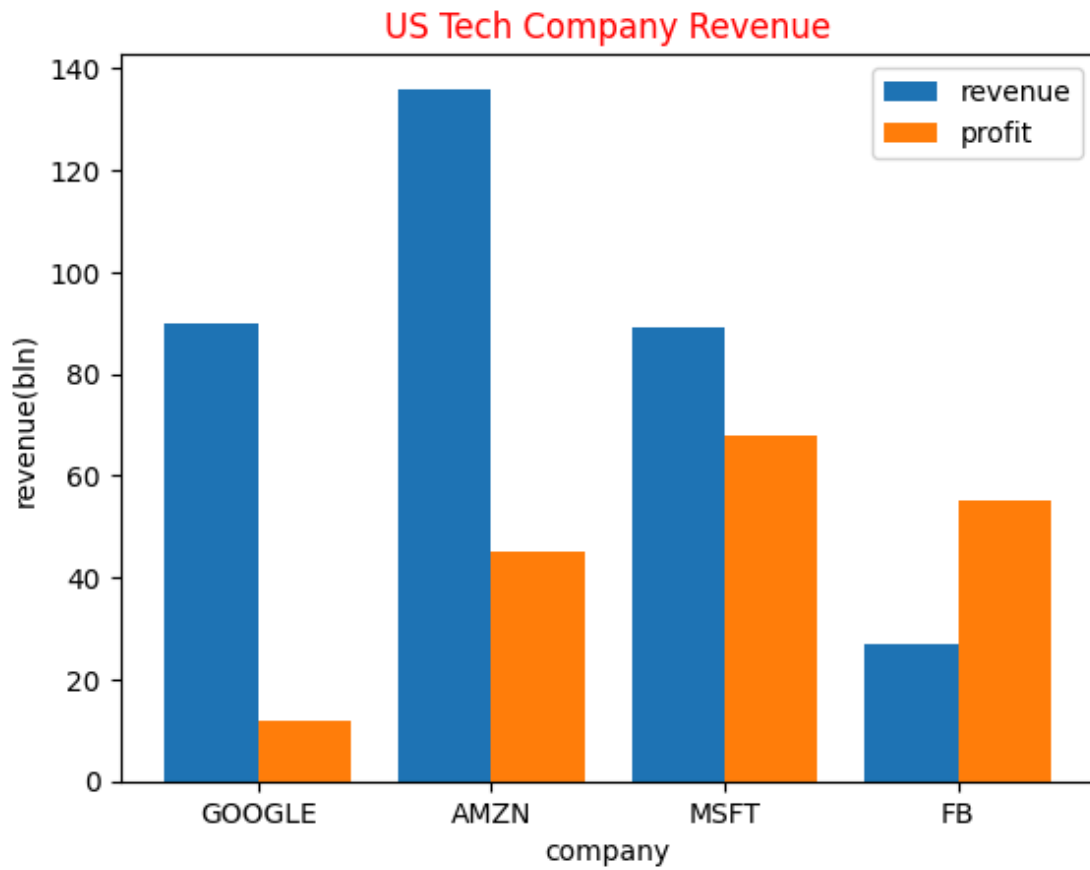


```
→ ([<matplotlib.axis.XTick at 0x7f48167a8550>,  
    <matplotlib.axis.XTick at 0x7f48167dc910>,  
    <matplotlib.axis.XTick at 0x7f4816a47590>,  
    <matplotlib.axis.XTick at 0x7f4816985050>],  
 [Text(0, 0, 'GOOGLE'),  
  Text(1, 0, 'AMZN'),  
  Text(2, 0, 'MSFT'),  
  Text(3, 0, 'FB')])
```



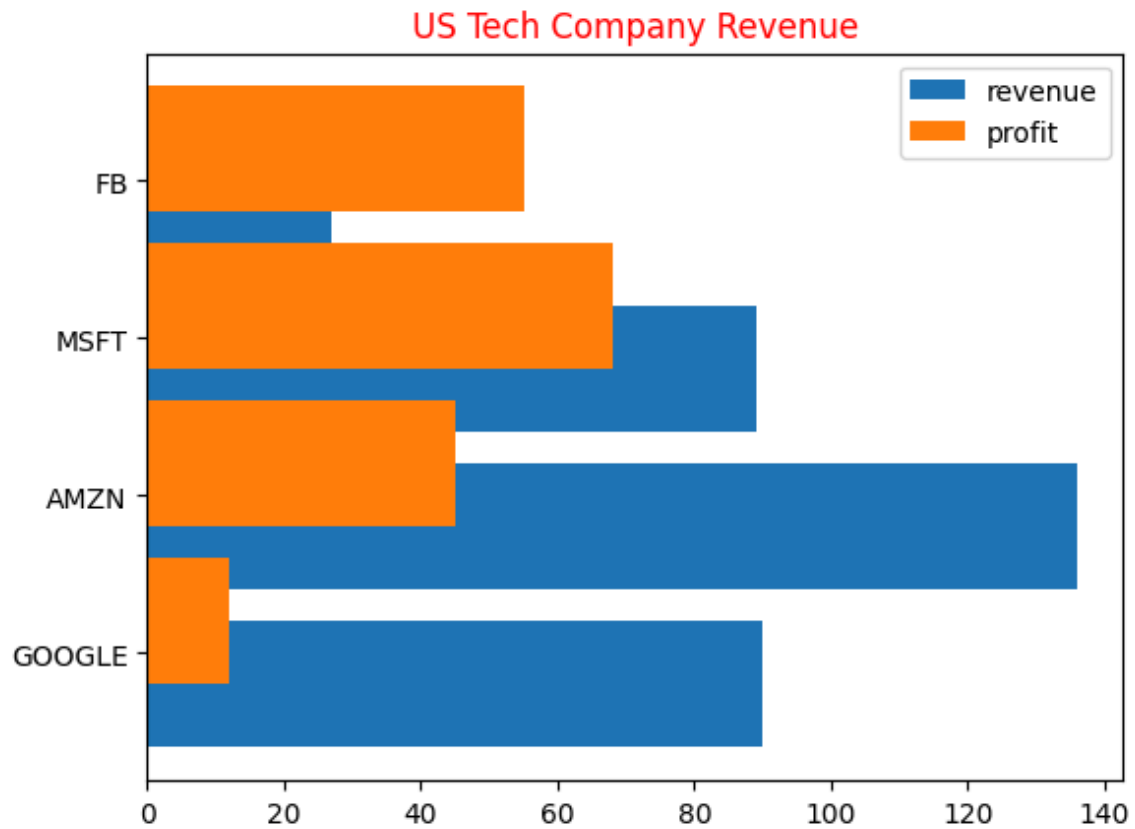
```
plt.xticks(x_pos,company)  
plt.xlabel("company")  
plt.ylabel("revenue(bln)")  
plt.title("US Tech Company Revenue",color="red")  
plt.bar(x_pos-0.2,revenue,width=0.4,label="revenue")  
plt.bar(x_pos+0.2, profit,width=0.4, label="profit")  
plt.legend()
```

↔ <matplotlib.legend.Legend at 0x7f48167058d0>



```
plt.yticks(x_pos,company)
plt.title("US Tech Company Revenue",color="red")
plt.barh(x_pos-0.2,revenue,label="revenue")
plt.barh(x_pos+0.2, profit, label="profit")
plt.legend()
```

↔ <matplotlib.legend.Legend at 0x7f481686d250>



Histograms

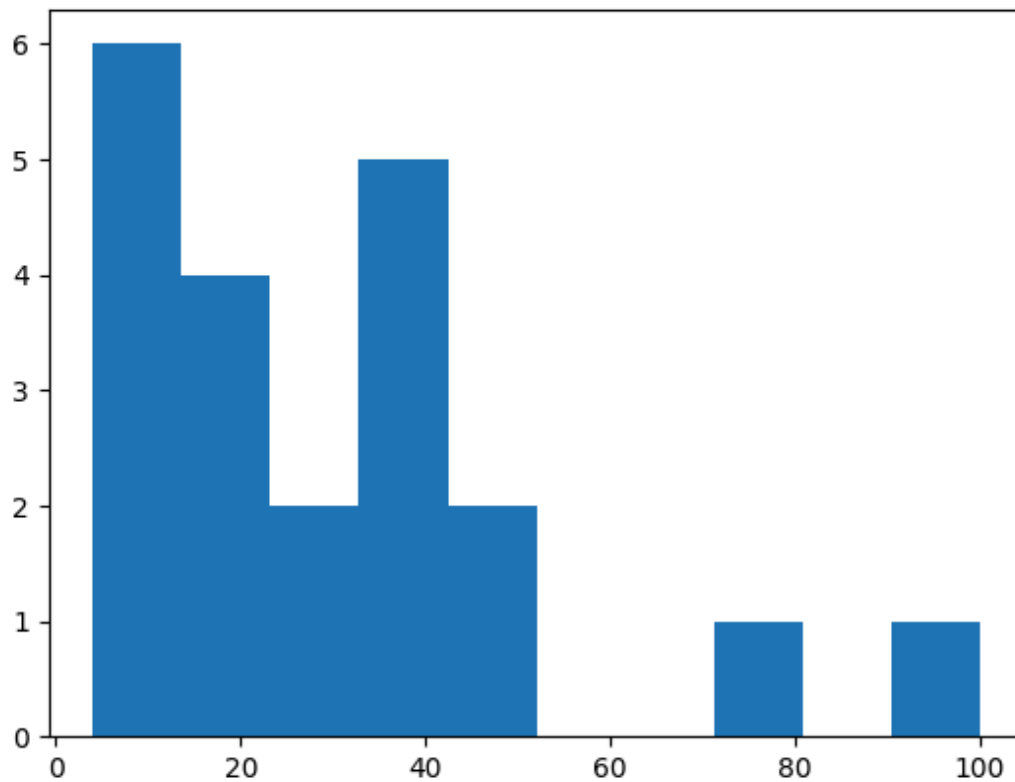
```
import matplotlib.pyplot as plt
%matplotlib inline
```

```
x = [21,22,23,4,5,6,77,8,9,10,31,32,33,34,35,36,37,18,49,50,100]
plt.hist(x,bins=3)
```

```

↳ (array([6., 4., 2., 5., 2., 0., 0., 1., 0., 1.]),
   array([ 4., 13.6, 23.2, 32.8, 42.4, 52., 61.6, 71.2, 80.8,
          90.4, 100. ]),
   <BarContainer object of 10 artists>)

```



```

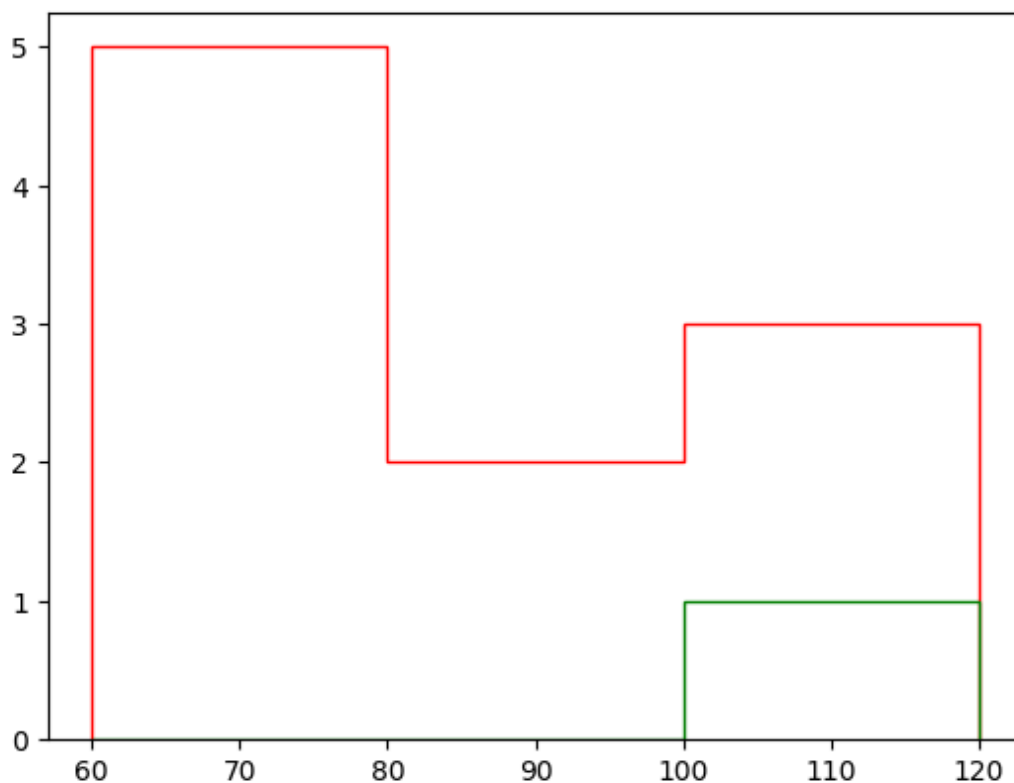
blood_sugar_men = [21,32,33,34,35,36,37,18,49,50,100]
blood_sugar_women = [68,79,70,89,90,100,115,69,120,59,78]
plt.hist([blood_sugar_men,blood_sugar_women], bins=[60,80,100,120], rwidth=0.9, color=["green","red"])

```

```

⇒ (array([[0., 0., 1.],
          [5., 2., 3.]]),
   array([ 60.,  80., 100., 120.]),
   <a list of 2 list[Polygon] objects>)

```

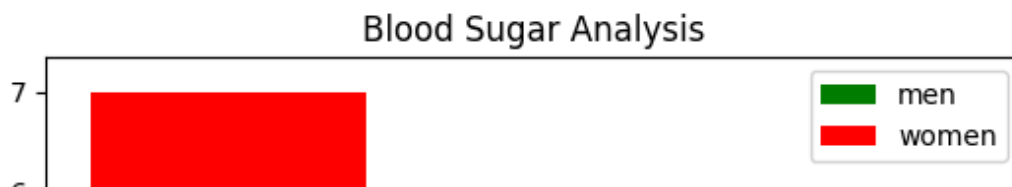


```

blood_sugar_men = [21,32,33,34,35,36,37,18,49,50,100,62]
blood_sugar_women = [68,74,73,70,86,82,100,110,69,120,59,78]
plt.xlabel("Blood Sugar")
plt.ylabel("Frequency")
plt.title("Blood Sugar Analysis")
plt.hist([blood_sugar_men,blood_sugar_women], bins=[60,80,100,120], rwidth=0.9, color=["green","red"],
         label=["men","women"], histtype="barstacked")
plt.legend()

```

 <matplotlib.legend.Legend at 0x7e1ed17afa50>



```
blood_sugar_men = [21,32,33,34,35,36,37,18,49,50,100,62]
blood_sugar_women = [68,74,73,70,86,82,100,110,69,120,59,78]
plt.xlabel("Blood Sugar")
plt.ylabel("Frequency")
plt.title("Blood Sugar Analysis")
plt.hist([blood_sugar_men,blood_sugar_women], bins=[60,80,100,120], rwidth=0.9, color=["green","red"],
        label=["men","women"], orientation='horizontal')
plt.legend()
```

 <matplotlib.legend.Legend at 0x7e1ece30f210>

